

Problem 1

Determine which of the following expressions are allowed in index notation, if they are not allowed state why (a , b , c , d , and e are arbitrary quantities) (10 points).

- (a) $a = b_i c_{ij} d_j$: allowed. i and j each appear twice and are summed, leaving no free index, consistent with the scalar a .
- (b) $a = b_i c_i + d_j$: not allowed. $b_i c_i$ has no free index while d_j carries free index j ; the two terms disagree.
- (c) $a_i = \delta_{ij} b_i + c_i$: not allowed. $\delta_{ij} b_i = b_j$ carries free index j , while c_i carries free index i ; the terms disagree.
- (d) $a_k = b_i c_{ki}$: allowed. i is summed, k is free and matches the left side.
- (e) $a_k = b_k c + d_i e_{ik}$: allowed. Both terms carry free index k , matching the left side.
- (f) $a_i = b_i + c_{ij} d_{ji} e_i$: not allowed. i appears three times in $c_{ij} d_{ji} e_i$.
- (g) $a_l = \epsilon_{ijk} b_j c_k$: not allowed. j and k are summed, leaving free index i on the right, but the left side carries free index l .
- (h) $a_{ij} = b_{ji}$: allowed. i and j are free on both sides.
- (i) $a_{ij} = b_i c_j + e_{jk}$: not allowed. The first term has free indices $\{i, j\}$, the second has $\{j, k\}$.
- (j) $a_{kl} = b_i c_{ki} d_l + e_{ki}$: not allowed. i is summed in the first term, leaving free indices $\{k, l\}$; e_{ki} carries free indices $\{k, i\}$, which do not match.

Problem 2

For three spatial dimensions, rewrite the following expressions in index notation and evaluate or simplify them using the values or parameters given, and the definitions of δ_{ij} and ϵ_{ijk} wherever possible. In parts b) through e), $\vec{x}$ is the position vector, with components x_i .

Part a)

$\vec{b} \cdot \vec{c}$, where $\vec{b} = (1, 4, 17)$ and $\vec{c} = (-4, -3, 1)$.
 $\vec{b} \cdot \vec{c} = b_i c_i$:

$$b_i c_i = (1)(-4) + (4)(-3) + (17)(1) = -4 - 12 + 17 = \boxed{1} \quad (1)$$

Part b)

$(\vec{u} \cdot \vec{\nabla}) \vec{x}$, where $\vec{u}$ = a vector with components u_i .

The j -th component is $u_i \partial_i x_j$, and $\partial x_j / \partial x_i = \delta_{ij}$. Thus

$$\left[(\vec{u} \cdot \vec{\nabla}) \vec{x} \right]_j = u_i \frac{\partial x_j}{\partial x_i} = u_i \delta_{ij} = u_j \quad (2)$$

so

$$\boxed{(\vec{u} \cdot \vec{\nabla}) \vec{x} = \vec{u}} \quad (3)$$

Part c)

$\vec{\nabla} \phi$, where $\phi = \vec{h} \cdot \vec{x}$ and $\vec{h}$ is a constant vector with components h_i .

Writing $\phi = h_i x_i$:

$$(\vec{\nabla} \phi)_j = \frac{\partial \phi}{\partial x_j} = \frac{\partial}{\partial x_j} (h_i x_i) = h_i \frac{\partial x_i}{\partial x_j} = h_i \delta_{ij} = h_j \quad (4)$$

so

$$\boxed{\vec{\nabla} \phi = \vec{h}} \quad (5)$$

Part d)

$\vec{\nabla} \times \vec{u}$, where $\vec{u} = \vec{\Omega} \times \vec{x}$ and $\vec{\Omega}$ is a constant vector with components Ω_i .

Write $u_p = \epsilon_{pqr} \Omega_q x_r$. Then

$$(\vec{\nabla} \times \vec{u})_i = \epsilon_{ijk} \partial_j u_k = \epsilon_{ijk} \partial_j (\epsilon_{klm} \Omega_l x_m) = \epsilon_{ijk} \epsilon_{klm} \Omega_l \frac{\partial x_m}{\partial x_j} = \epsilon_{ijk} \epsilon_{klm} \Omega_l \delta_{jm}$$

Setting $m = j$:

$$(\vec{\nabla} \times \vec{u})_i = \epsilon_{ijk} \epsilon_{klj} \Omega_l = \epsilon_{kij} \epsilon_{klj} \Omega_l$$

using $\epsilon_{ijk} = \epsilon_{kij}$. Applying $\epsilon_{kij}\epsilon_{klj} = \delta_{il}\delta_{jj} - \delta_{ij}\delta_{jl}$ and summing over j ($\delta_{jj} = 3$, $\delta_{ij}\delta_{jl} = \delta_{il}$):

$$(\vec{\nabla} \times \vec{u})_i = \Omega_l (3\delta_{il} - \delta_{il}) = 2\delta_{il}\Omega_l = 2\Omega_i$$

so

$$\boxed{\vec{\nabla} \times \vec{u} = 2\vec{\Omega}} \quad (6)$$

Part e)

$\overleftrightarrow{C} \cdot \vec{x}$, where

$$\overleftrightarrow{C} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$$

$$(\overleftrightarrow{C} \cdot \vec{x})_i = C_{ij}x_j:$$

$$\boxed{\overleftrightarrow{C} \cdot \vec{x} = \begin{pmatrix} x_1 + 2x_2 + 3x_3 \\ x_2 + 2x_3 \\ x_3 \end{pmatrix}} \quad (7)$$

Problem 3

Prove that the following expression is true using Cartesian index notation without the use of vector identities (if you want to use a vector identity you must first show it is true using index notation).

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{F}) = \vec{\nabla} (\vec{\nabla} \cdot \vec{F}) - \nabla^2 \vec{F}$$

Using the cyclic property of the permutation symbol from MATH 321 and lecture, $\epsilon_{ijk} = \epsilon_{kij}$, along with the ϵ - δ identity

$$\epsilon_{kij}\epsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}, \quad (8)$$

the curl of the curl becomes

$$\begin{aligned} \left[\vec{\nabla} \times (\vec{\nabla} \times \vec{F}) \right]_i &= \epsilon_{ijk} \partial_j (\epsilon_{klm} \partial_l F_m) \\ &= \epsilon_{ijk} \epsilon_{klm} \partial_j \partial_l F_m \\ &= \epsilon_{kij} \epsilon_{klm} \partial_j \partial_l F_m \end{aligned}$$

Applying this property,

$$\begin{aligned} \left[\vec{\nabla} \times (\vec{\nabla} \times \vec{F}) \right]_i &= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}) \partial_j \partial_l F_m \\ &= \delta_{il}\delta_{jm} \partial_j \partial_l F_m - \delta_{im}\delta_{jl} \partial_j \partial_l F_m \end{aligned}$$

The first term sets $l = i, m = j$: $\partial_j \partial_i F_j = \partial_i (\partial_j F_j) = \partial_i (\vec{\nabla} \cdot \vec{F})$. The second sets $m = i, l = j$: $\partial_j \partial_j F_i = \nabla^2 F_i$. This gives

$$\left[\vec{\nabla} \times (\vec{\nabla} \times \vec{F}) \right]_i = \partial_i (\vec{\nabla} \cdot \vec{F}) - \nabla^2 F_i = \left[\vec{\nabla} (\vec{\nabla} \cdot \vec{F}) - \nabla^2 \vec{F} \right]_i$$

for $i = 1, 2, 3$, so

$$\boxed{\vec{\nabla} \times (\vec{\nabla} \times \vec{F}) = \vec{\nabla} (\vec{\nabla} \cdot \vec{F}) - \nabla^2 \vec{F}} \quad (9)$$

□

Problem 4

Prove that the following expressions are true using Cartesian index notation without the use of vector identities (if you want to use a vector identity you must first show it is true using index notation).

$$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{c} \cdot (\vec{d} \times \vec{a})] \vec{b} - [\vec{c} \cdot (\vec{d} \times \vec{b})] \vec{a}$$

And

$$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{a} \cdot (\vec{b} \times \vec{d})] \vec{c} - [\vec{a} \cdot (\vec{b} \times \vec{c})] \vec{d}$$

Both follow from two lemmas, proved below in index notation for arbitrary vectors $\vec{p}, \vec{q}, \vec{r}$.

Lemma A: $\vec{p} \times (\vec{q} \times \vec{r}) = \vec{q}(\vec{p} \cdot \vec{r}) - \vec{r}(\vec{p} \cdot \vec{q})$

Using $\epsilon_{ijk} = \epsilon_{kij}$ and $\epsilon_{kij}\epsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$ from Problem 3:

$$\begin{aligned} [\vec{p} \times (\vec{q} \times \vec{r})]_i &= \epsilon_{ijk} p_j (\epsilon_{klm} q_l r_m) = \epsilon_{ijk} \epsilon_{klm} p_j q_l r_m = \epsilon_{kij} \epsilon_{klm} p_j q_l r_m \\ &= (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) p_j q_l r_m \\ &= \delta_{il} \delta_{jm} p_j q_l r_m - \delta_{im} \delta_{jl} p_j q_l r_m \\ &= q_i (p_j r_j) - r_i (p_j q_j) = q_i (\vec{p} \cdot \vec{r}) - r_i (\vec{p} \cdot \vec{q}) \end{aligned}$$

so $\vec{p} \times (\vec{q} \times \vec{r}) = \vec{q}(\vec{p} \cdot \vec{r}) - \vec{r}(\vec{p} \cdot \vec{q})$. ■

Lemma B: $(\vec{p} \times \vec{q}) \cdot \vec{r} = \vec{p} \cdot (\vec{q} \times \vec{r})$

In index notation,

$$(\vec{p} \times \vec{q}) \cdot \vec{r} = \epsilon_{iab} p_a q_b r_i, \quad \vec{p} \cdot (\vec{q} \times \vec{r}) = \epsilon_{iab} p_i q_a r_b$$

Renaming the dummy indices $(i, a, b) \rightarrow (b, i, a)$ in the first expression gives

$$\epsilon_{iab} p_a q_b r_i = \epsilon_{bia} p_i q_a r_b = \epsilon_{iab} p_i q_a r_b$$

using $\epsilon_{bia} = \epsilon_{iab}$. Hence $(\vec{p} \times \vec{q}) \cdot \vec{r} = \vec{p} \cdot (\vec{q} \times \vec{r})$. ■

The antisymmetry of the cross product follows the same way: $(\vec{q} \times \vec{p})_i = \epsilon_{ijk} q_j p_k$; renaming $j \leftrightarrow k$ gives $\epsilon_{ikj} q_k p_j = \epsilon_{ikj} p_j q_k = -\epsilon_{ijk} p_j q_k = -(\vec{p} \times \vec{q})_i$. So $\vec{q} \times \vec{p} = -\vec{p} \times \vec{q}$.

Proof of the second identity

Apply Lemma A with $\vec{p} = \vec{a} \times \vec{b}$, $\vec{q} = \vec{c}$, $\vec{r} = \vec{d}$:

$$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \vec{c} [(\vec{a} \times \vec{b}) \cdot \vec{d}] - \vec{d} [(\vec{a} \times \vec{b}) \cdot \vec{c}]$$

By Lemma B, $(\vec{a} \times \vec{b}) \cdot \vec{d} = \vec{a} \cdot (\vec{b} \times \vec{d})$ and $(\vec{a} \times \vec{b}) \cdot \vec{c} = \vec{a} \cdot (\vec{b} \times \vec{c})$. Substituting,

$$\boxed{(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{a} \cdot (\vec{b} \times \vec{d})] \vec{c} - [\vec{a} \cdot (\vec{b} \times \vec{c})] \vec{d}} \quad (10)$$

Proof of the first identity

Apply Lemma A with $\vec{p} = \vec{c} \times \vec{d}$, $\vec{q} = \vec{a}$, $\vec{r} = \vec{b}$:

$$(\vec{c} \times \vec{d}) \times (\vec{a} \times \vec{b}) = \vec{a} \left[(\vec{c} \times \vec{d}) \cdot \vec{b} \right] - \vec{b} \left[(\vec{c} \times \vec{d}) \cdot \vec{a} \right]$$

By antisymmetry, $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = -(\vec{c} \times \vec{d}) \times (\vec{a} \times \vec{b})$, so

$$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \vec{b} \left[(\vec{c} \times \vec{d}) \cdot \vec{a} \right] - \vec{a} \left[(\vec{c} \times \vec{d}) \cdot \vec{b} \right]$$

By Lemma B, $(\vec{c} \times \vec{d}) \cdot \vec{a} = \vec{c} \cdot (\vec{d} \times \vec{a})$ and $(\vec{c} \times \vec{d}) \cdot \vec{b} = \vec{c} \cdot (\vec{d} \times \vec{b})$. Substituting,

$$\boxed{(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \left[\vec{c} \cdot (\vec{d} \times \vec{a}) \right] \vec{b} - \left[\vec{c} \cdot (\vec{d} \times \vec{b}) \right] \vec{a}} \quad (11)$$

□

Appendix: Code and Handwritten Derivations

All supporting derivations for this homework are available in the GitHub repository:

<https://github.com/tjjones6/ME563/tree/main/HW2>